

# AWS Certificate Manager (ACM) + CloudFront - Lab Guide

## Objective

Use **AWS Certificate Manager (ACM)** to create a **public SSL/TLS certificate**, and attach it to a **CloudFront distribution** to serve a static website over HTTPS.

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## Prerequisites

- Registered domain (e.g., example.com)
  - Access to domain's DNS records (preferably Route 53)
  - An S3 bucket with a static website hosted
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## Step 1: Create and Upload Static Website to S3

1. Go to **S3 Console** → Click **Create bucket**
  - Bucket name: `my-website-bucket`
  - Disable "Block all public access"
  - Enable **Static website hosting**
  - Index document: `index.html`

2. Upload `index.html` with basic content:

```
<html>
<head><title>My Website</title></head>
<body><h1>Welcome to my HTTPS website!</h1></body>
</html>
```

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## Step 2: Request Public Certificate from ACM

1. Go to **AWS Certificate Manager (ACM)** → Click **Request a certificate**
  2. Choose **Request a public certificate**
  3. Enter your domain name (e.g., `www.example.com`)
  4. Choose **DNS Validation**
  5. Confirm and request the certificate
  6. In the validation screen:
    - Copy the **CNAME record**
    - Add this record to your domain's DNS provider (or Route 53)
  7. Wait until the certificate status becomes **"Issued"**
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## Step 3: Set Up CloudFront Distribution

1. Go to **CloudFront** → Click **Create Distribution**

## 2. Origin Settings:

- Origin domain: Select your S3 bucket
- Origin access: Public (or use OAC if configured)

## 3. Default Cache Behavior:

- Viewer protocol policy: **Redirect HTTP to HTTPS**
- Allowed HTTP methods: GET, HEAD

## 4. Settings:

- Alternate domain name (CNAME): [www.example.com](http://www.example.com)
- SSL certificate: Select the **ACM certificate** you created
- Price class: Use default (or limit to specific edge locations)

## 5. Click **Create Distribution**

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## Step 4: Configure DNS

1. Go to **Route 53** or your domain registrar
  2. Add a **CNAME** or **A Record (alias)** for [www.example.com](http://www.example.com) pointing to your CloudFront distribution domain name (e.g., [d1234abcd.cloudfront.net](http://d1234abcd.cloudfront.net))
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## Step 5: Test the Website

- Open a browser and go to: <https://www.example.com>
  - You should see the secure padlock 
  - The website content should load via HTTPS from CloudFront
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## Cleanup

To avoid ongoing costs:

- Delete the CloudFront distribution (takes ~15 mins)
  - Delete the ACM certificate (optional)
  - Remove DNS records
  - Delete S3 bucket and contents
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## Notes

- ACM public certs are **free** and automatically renew.
  - CloudFront supports only **ACM certificates in the US East (N. Virginia) region**.
  - DNS validation is recommended for automation and ease of renewal.
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## Extra Exercises

- Add **custom error pages** to CloudFront.
- Enable **access logging** for CloudFront.
- Use **WAF (Web Application Firewall)** for additional security.
- Automate the setup using **CloudFormation** or **Terraform**.